Review: Foundations and unconstrained optimization

MA3236 Non-Linear Programming (L1-L11)

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Short review

Lectures	Skills to practise
L01–L05	Describe models; establish existence and convexity; compute derivatives.
L06–L07	Find every stationary point; distinguish local tests from global certificates.
L08–L11	Perform iterations; state assumptions and interpret convergence claims.

Focus. The main calculations or reasonings to master.

Mistake. Making invalid inferences or calculation errors.

Problem formulation

L01: models and notation

$$\min_{\mathbf{x}\in\mathcal{S}} f(\mathbf{x}), \qquad \mathcal{S} = \{\mathbf{x} \in \mathcal{D} : g_i(\mathbf{x}) = 0,\ h_j(\mathbf{x}) \le 0\}.$$

- Identify the **objective, variables, constraints, feasible set**.
- Check whether $\mathcal{S}$ is **nonempty** before seeking an optimizer.

$$\max_{\mathbf{x}\in\mathcal{S}} f(\mathbf{x}) = -\min_{\mathbf{x}\in\mathcal{S}} [-f(\mathbf{x})].$$

Focus. Concepts of objective, variables, constraint, and feasible set.

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Focus. Concepts of objective, variables, constraint, and feasible set.

Mistake. Applying canonical-form results without adjusting objective or constraint signs.

Existence and attainment

L01–L02: well-posedness

A global minimizer exists if either of the following applies:

- (**Weierstrass**) $\mathcal{S} \neq \varnothing$ **compact** + f continuous on $\mathcal{S}$.
- $\mathcal{S} \neq \varnothing$ closed + f continuous, **coercive** ($f(\mathbf{x}) \to +\infty$ as $\|\mathbf{x}\| \to \infty$).

Problem	Infimum	Attained?
$\min_{x \in \mathbf{R}} x^2$	0	Yes, at 0
$\inf_{0<x<1} x$	0	No
$\inf_{x \in \mathbf{R}} e^x$	0	No

Focus. Weierstrass extreme-value theorem, concept of coercivity.

Existence and attainment

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$\inf_{x \in \mathbf{R}} e^x$	0	No

Focus. Weierstrass extreme-value theorem, concept of coercivity.

Mistake. (1) Invoking existence without its hypotheses. (2) Equating a finite infimum with attainment. (3) Inferring an unbounded objective from an unbounded feasible set.

Local, global, strict, and unique

L02–L04: definitions and their implications

For a feasible $\mathbf{x}^*$, compare $f(\mathbf{x}^*)$ with $f(\mathbf{x})$:

Type	Inequality	Points compared
Local minimum	$f(\mathbf{x}^*) \leq f(\mathbf{x})$	$\mathbf{x} \in \mathcal{S}$ close to $\mathbf{x}^*$
Strict local minimum	$f(\mathbf{x}^*) < f(\mathbf{x})$	$\mathbf{x}^* \neq \mathbf{x} \in \mathcal{S}$ close to $\mathbf{x}^*$
Global minimum	$f(\mathbf{x}^*) \leq f(\mathbf{x})$	every $\mathbf{x} \in \mathcal{S}$

- Global optimality $\Rightarrow$ local optimality.
- **Convex** f on convex $\mathcal{S}$: local $\Rightarrow$ global.
- Strictly convex f on convex $\mathcal{S}$ $\Rightarrow$ **at most** one global minimizer.

Focus. Concepts of (strict) local and global optimalities.

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Focus. Concepts of (strict) local and global optimalities.

Mistake. Inferring existence from (strict) convexity alone.

Set convexity

$$\mathbf{x}, \mathbf{y} \in C, \;\; t \in [0,1] \quad \Longrightarrow \quad (1-t)\mathbf{x} + t\mathbf{y} \in C.$$

- Convex combination:

$$\mathbf{x}_i \in C, \;\; t_i \in [0,1], \; \sum_i t_i = 1 \quad \Longrightarrow \quad \sum_i t_i \mathbf{x}_i \in C.$$

- **Intersections** of convex sets are convex.
- Affine equalities and inequalities define convex sets.
- Each **sublevel set** of a convex function is convex.

Focus. Concept of set convexity, convexity-preserving operations.

Set convexity

L03: convex sets

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- Each **sublevel set** of a convex function is convex.

Focus. Concept of set convexity, convexity-preserving operations.

Mistake. (1) Testing only selected points or weights. (2) Treating arbitrary unions as convex. (3) Inferring a convex equality set from convex constraint functions.

Convex functions

L03: convexity without differentiation

On a convex domain $\mathcal{D}$,

$$f((1-t)\mathbf{x}+t\mathbf{y}) \le (1-t)f(\mathbf{x})+tf(\mathbf{y}), \quad \forall\, \mathbf{x},\mathbf{y}\in\mathcal{D}, \quad 0\le t\le 1.$$

- **Jensen's inequality**:

$$f(\textstyle\sum_i t_i\mathbf{x}_i) \le \sum_i t_i f(\mathbf{x}_i), \quad \mathbf{x}_i\in\mathcal{D}, \quad t_i\in[0,1], \quad \sum_i t_i = 1.$$

- **Nonnegative weighted sums** preserve convexity.
- **Pointwise maxima** preserve convexity.
- **Composition**: ϕ convex, nondecreasing + f convex $\Rightarrow \phi\circ f$ convex.

Focus. Concept of function convexity, convexity-preserving operations.

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- **Composition**: ϕ convex, nondecreasing + f convex $\Rightarrow \phi\circ f$ convex.

Focus. Concept of function convexity, convexity-preserving operations.

Mistake. (1) Allowing negative weights in a convex sum. (2) Treating pointwise minima as convexity-preserving. (3) Dropping the monotonicity condition in composition.

Derivatives

L04–L05: gradients, Hessians, and Taylor expansion

For $f \in C^2$, write $\mathbf{H}_f = \nabla^2 f$. Then

$$f(\mathbf{x}+\mathbf{s}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{s} + \tfrac{1}{2}\mathbf{s}^T \mathbf{H}_f(\mathbf{x})\mathbf{s} + o(\|\mathbf{s}\|^2).$$

For a real symmetric $\mathbf{Q}$,

$$f(\mathbf{x}) = \tfrac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x} \quad \Longrightarrow \quad \nabla f = \mathbf{Q}\mathbf{x} - \mathbf{c}, \quad \mathbf{H}_f = \mathbf{Q}.$$

Focus. Concepts and calculations of gradient and Hessian.

Derivatives

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Focus. Concepts and calculations of gradient and Hessian.

Mistake. (1) Losing cross terms or factors of two.
(2) Using $\nabla(\frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x}) = \mathbf{Q}\mathbf{x}$ without symmetry.

Three equivalent tests for function convexity

L03–L05: convexity with differentiation

Let $\mathcal{D}$ be open and convex. For $f \in C^2(\mathcal{D})$, the following are equivalent:

1. f lies below every chord.
2. **Every tangent plane** is a global lower bound:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \quad (\forall\ \mathbf{x}, \mathbf{y} \in \mathcal{D}).$$

3. (**Most recommended** if possible) $\mathbf{H}_f(\mathbf{x}) \succeq 0$ at **every** $\mathbf{x} \in \mathcal{D}$.

Focus. Relation between convexity and derivatives (esp., curvature).

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Focus. Relation between convexity and derivatives (esp., curvature).

Mistake. (1) Checking the Hessian at only one point. (2) Inferring a PD Hessian everywhere from strict convexity.

(Semi-)definiteness

L05: real symmetric matrices

Property	Eigenvalues	Equivalent quadratic-form test
PD	All > 0	$\mathbf{d}^T H\mathbf{d} > 0$ for $\mathbf{d} \neq \mathbf{0}$
PSD	All ≥ 0	$\mathbf{d}^T H\mathbf{d} \geq 0$ for every $\mathbf{d}$
Indefinite	Positive and negative	Both signs occur

- PD: all **leading** principal minors are positive (**Sylvester's criterion**).
- PSD: all principal minors are nonnegative.
- **ND/NSD**: apply the PD/PSD test to $-H$.

Focus. Computation of eigenvalues, Sylvester's criterion.

(Semi-)definiteness

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- PSD: all principal minors are nonnegative.
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Focus. Computation of eigenvalues, Sylvester's criterion.

Mistake. (1) Checking only leading minors for PSD. (2) Allowing zero eigenvalues in an indefinite matrix. (3) Using the PD/PSD sign tests unchanged for ND/NSD.

Unconstrained optimality

L06–L07: assume an interior point and $f \in C^2$

Any $\mathbf{x}$ such that $\nabla f(\mathbf{x}) = \mathbf{0}$ is a **stationary point**.

$$
\begin{aligned}
[\mathbf{N.}]: \quad & \mathbf{x}^* \text{ local minimum} && \Longrightarrow && \nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \succeq 0. \\
[\mathbf{S.}]: \quad & \nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \succ 0 && \Longrightarrow && \mathbf{x}^* \text{ strict local minimum.} \\
& \nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \text{ indefinite} && \Longrightarrow && \mathbf{x}^* \text{ saddle.}
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- A singular semidefinite Hessian leaves the test inconclusive.

Focus. Necessity vs sufficiency, possible roles of stationarity (next page).

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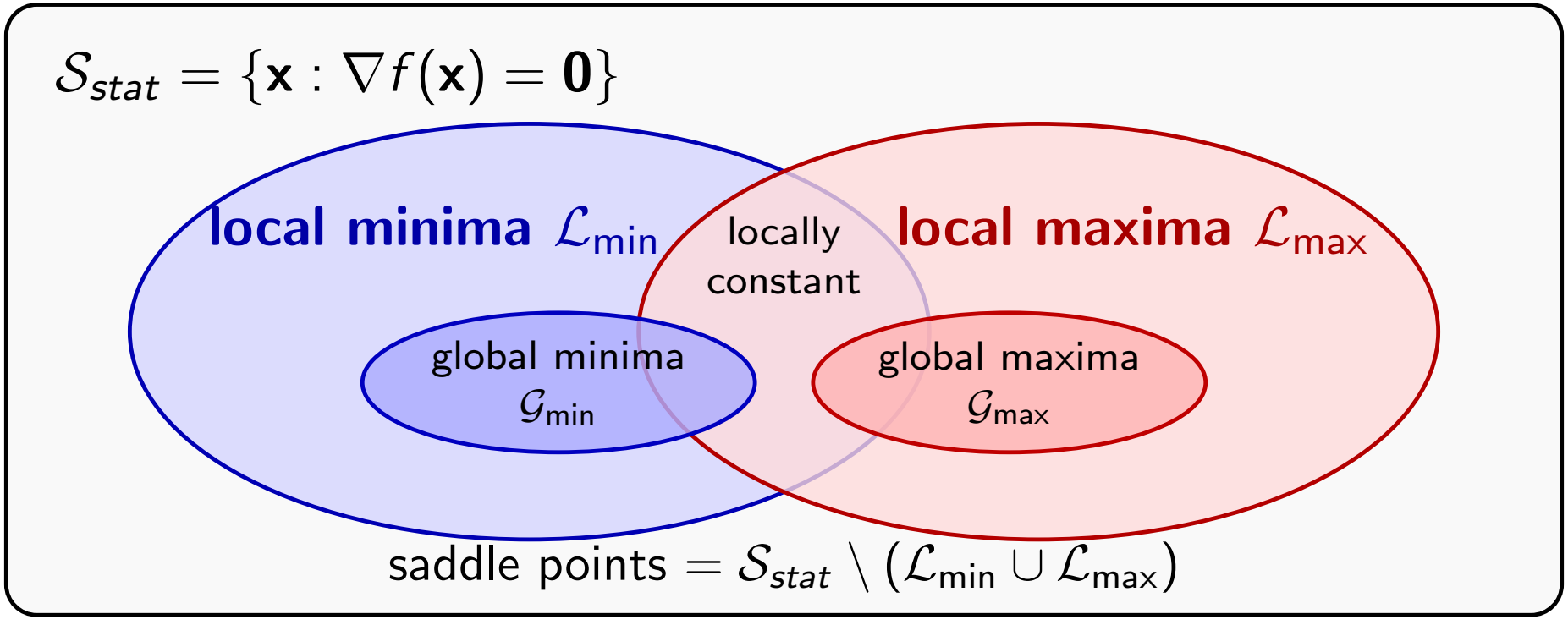
Focus. Necessity vs sufficiency, possible roles of stationarity (next page).

Mistake. (1) Reversing necessary or sufficient conditions. (2) Promoting a local minimum to a global one without further assumptions. (3) Inferring well-posedness from existence of a stationary point.

The possible roles of a stationary point

L06–L07: differentiable f on an open domain $\mathcal{D}$

Suppose $\mathcal{G}_{\min} \cap \mathcal{G}_{\max} = \emptyset$:



$$\mathcal{G}_{\min} \subseteq \mathcal{L}_{\min} \subseteq \mathcal{S}_{stat}, \qquad \mathcal{G}_{\max} \subseteq \mathcal{L}_{\max} \subseteq \mathcal{S}_{stat}.$$

If $\mathcal{G}_{\min} \cap \mathcal{G}_{\max} \neq \varnothing$, f is constant and all five sets equal $\mathcal{D}$.

Complete quadratic classification

L07: do not infer more than the Hessian says

For an **exact quadratic**

$$q(\mathbf{x}) = \tfrac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}, \quad \mathbf{Q} = \mathbf{Q}^T$$

Condition	Conclusion
$\mathbf{Q} \succ 0$	Unique minimizer $\mathbf{Q}^{-1}\mathbf{c}$
$\mathbf{Q} \succeq 0$, $\mathbf{c} \in \text{Range}(\mathbf{Q})$	Minimizers $\mathbf{x}_0 + \text{Null}(\mathbf{Q})$, where $\mathbf{Q}\mathbf{x}_0 = \mathbf{c}$
$\mathbf{Q} \succeq 0$, $\mathbf{c} \notin \text{Range}(\mathbf{Q})$	Unbounded below along a null direction
$\mathbf{Q}$ has a negative eigval	Unbounded below along negative curvature

Focus. The gradient and Hessian determine the quadratic classification.

Complete quadratic classification

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Focus. The gradient and Hessian determine the quadratic classification.

Mistake. Applying the complete quadratic classification to a general nonlinear objective using its Hessian at one point.

(Abstract) convex first-order optimality

L04: a global certificate from convexity

For a feasible $\mathbf{x}^* \in C$, differentiable convex f, and convex C,

$$\mathbf{x}^* \in \arg\min_{C} f \quad \Longleftrightarrow \quad \nabla f(\mathbf{x}^*)^T(\mathbf{y} - \mathbf{x}^*) \geq 0 \quad \forall\, \mathbf{y} \in C.$$

Here f is differentiable on an open neighborhood of C.

- At an **interior point** of $C \Rightarrow \nabla f(\mathbf{x}^*) = \mathbf{0}$.
- At a **boundary point**, only feasible directions matter.

Focus. Convexity $\Rightarrow$ the first-order inequality is necessary and sufficient.

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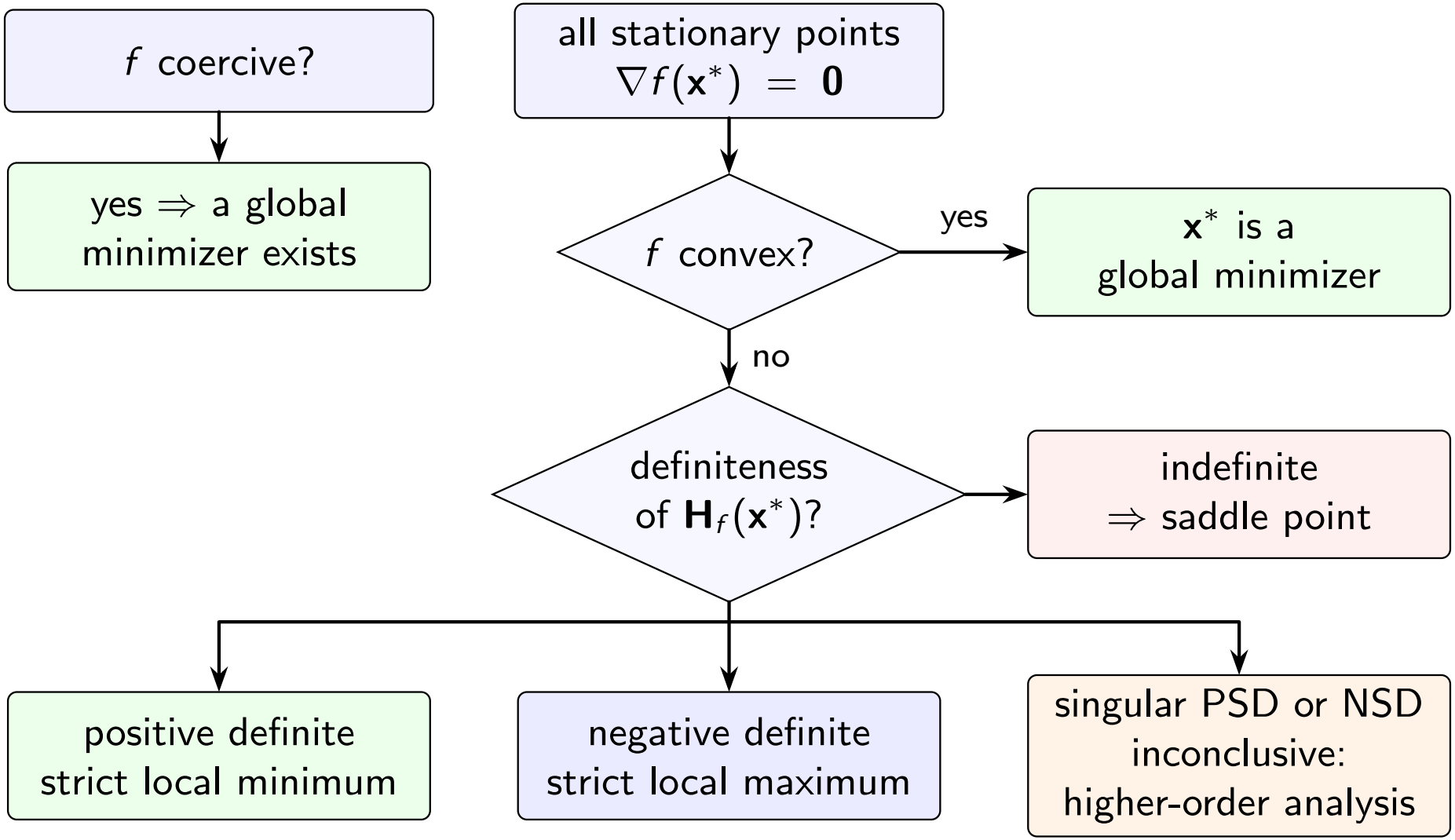
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Focus. Convexity $\Rightarrow$ the first-order inequality is necessary and sufficient.

Mistake. Always requiring $\nabla f = \mathbf{0}$ at a boundary minimizer.

Unconstrained optimality workflow

L06–L07: $\min_{\mathbf{x}\in\mathbf{R}^n} f(\mathbf{x})$, with $f \in C^2$



Global comparison. Once attainment is established, minimize f over the *entire* stationary set. If it is finite, compare all values, including candidates with singular Hessians.

Iterations, stopping, and convergence claims

L08: line-search framework

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)}\mathbf{d}^{(k)}, \qquad \varphi^{(k)}(\alpha) := f(\mathbf{x}^{(k)} + \alpha\mathbf{d}^{(k)}).$$

- **Descent direction**: $\mathbf{d}^{(k)}$ is usually chosen to satisfy:

$$(\varphi^{(k)})'(0) = (\nabla f(\mathbf{x}^{(k)}))^T\mathbf{d}^{(k)} < 0 \quad \text{if} \quad \mathbf{g}^{(k)} \neq \mathbf{0}.$$

- **Step size**: one-dim. methods; **Armijo backtracking**: $\alpha \leftarrow \beta\alpha$ until **Armijo sufficient-decrease condition** holds

$$f(\mathbf{x}^{(k)} + \alpha\mathbf{d}^{(k)}) \leq f(\mathbf{x}^{(k)}) + \sigma\alpha(\nabla f(\mathbf{x}^{(k)}))^T\mathbf{d}^{(k)}, \quad 0 < \sigma, \beta < 1.$$

Focus. Descent direction, Armijo backtracking, Armijo sufficient-decrease condition.

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Focus. Descent direction, Armijo backtracking, Armijo sufficient-decrease condition.

Mistake. Performing Armijo backtracking with a non-descent direction.

Iterations, stopping, and convergence claims (cont'd)

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)}\mathbf{d}^{(k)}, \qquad \varphi^{(k)}(\alpha) := f(\mathbf{x}^{(k)} + \alpha\mathbf{d}^{(k)}).$$

- **Stopping**: absolute/scaled residual based on, e.g., $\|\nabla f(\mathbf{x}^{(k)})\|$, and maximum iteration number/running time, ...
- (Asymptotic) **convergence rate**: for a specified error $e_k \to 0$:

  $$e_{k+1} \le c e_k \ (0 < c < 1): \ \textbf{Q-linear}; \quad e_{k+1} \le C e_k^2: \ \textbf{Q-quadratic},$$

  with these bounds holding for all sufficiently large k.
- **Complexity**: iterations, function/gradient/Hessian evaluations, ...

Focus. Aim of algs.: $\|\nabla f\| \approx 0$ (in this course), convergence rate, complexity to a given tolerance.

Iterations, stopping, and convergence claims (cont'd)

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Focus. Aim of algs.: $\|\nabla f\| \approx 0$ (in this course), convergence rate, complexity to a given tolerance.

Mistake. (1) Inferring optimality from a small $\|\nabla f\|$. (2) Claiming a convergence rate without hypotheses or specifying the error.

One-dimensional methods

L08: three basic choices, used in the line-search framework

Let $\varphi = \varphi^{(k)}$.

Method	Required info./setting	Update or guarantee	Reductions / rate
Bisection	Continuous φ'; $\varphi'(a) < 0 < \varphi'(b)$	Keep the half with a sign change	$N \geq \lceil \log_2(\ell_0/\varepsilon) \rceil$
Newton	φ', φ''; $\varphi''(x) \neq 0$	$x^+ = x - \dfrac{\varphi'(x)}{\varphi''(x)}$	Local Q-quadratic*
Golden section	φ; strict unimodality on $[a, b]$	Reuse one value; shrink by $\rho \approx 0.618$	$N \geq \left\lceil \dfrac{\log(\ell_0/\varepsilon)}{\log(1/\rho)} \right\rceil$

$\ell_0 = b - a$, $0 < \varepsilon < \ell_0$: target bracket length; N: number of reductions.
*Start close to x^*, with $\varphi'(x^*) = 0$, $\varphi''(x^*) \neq 0$, and φ'' locally Lipschitz.

Focus. Principles behind bisection, golden section (bracketing) and Newton (Taylor expansion).

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Focus. Principles behind bisection, golden section (bracketing) and Newton (Taylor expansion).

Mistake. (1) Treating every root of φ' as a minimum. (2) Applying a rate without the stated hypotheses.

Multi-dimensional methods: Newton's method

L09: Newton from a local quadratic model

At $\mathbf{x}^{(k)}$, let $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$ and $\mathbf{H}_f^{(k)} = \mathbf{H}_f(\mathbf{x}^{(k)})$. Consider

$$m_{\mathrm{N}}^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T\mathbf{d} + \tfrac{1}{2}\mathbf{d}^T\mathbf{H}_f^{(k)}\mathbf{d}.$$

Its stationarity **(Newton) equation** is

$$\mathbf{H}_f^{(k)}\mathbf{d}^{(k)} = -\mathbf{g}^{(k)} \quad \Longrightarrow \quad \text{Newton iteration : } \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{d}^{(k)}.$$

- $\mathbf{H}_f^{(k)} \succ 0$: the direction uniquely minimizes the model and
$$\mathbf{g}^{(k)} \neq \mathbf{0} \quad \Longrightarrow \quad (\mathbf{g}^{(k)})^T\mathbf{d}^{(k)} < 0 \text{ (descent direction).}$$
- $\mathbf{H}_f^{(k)}$ nonsingular, indefinite: unique $\mathbf{d}^{(k)}$, but not model minimum.
- $\mathbf{H}_f^{(k)}$ singular: check consistency and curvature.

Focus. Newton equation and its solution.

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$$m_{\mathrm{N}}^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T\mathbf{d} + \tfrac{1}{2}\mathbf{d}^T\mathbf{H}_f^{(k)}\mathbf{d}.$$

Its stationarity **(Newton) equation** is

$$\mathbf{H}_f^{(k)}\mathbf{d}^{(k)} = -\mathbf{g}^{(k)} \quad \Longrightarrow \quad \text{Newton iteration : } \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{d}^{(k)}.$$

- $\mathbf{H}_f^{(k)} \succ 0$: the direction uniquely minimizes the model and
$$\mathbf{g}^{(k)} \neq \mathbf{0} \quad \Longrightarrow \quad (\mathbf{g}^{(k)})^T\mathbf{d}^{(k)} < 0 \text{ (descent direction).}$$
- $\mathbf{H}_f^{(k)}$ nonsingular, indefinite: unique $\mathbf{d}^{(k)}$, but not model minimum.
- $\mathbf{H}_f^{(k)}$ singular: check consistency and curvature.

Focus. Newton equation and its solution.

Mistake. (1) Assuming the Newton equation always has a unique solution. (2) Dropping the minus sign on $\mathbf{g}^{(k)}$.

Multi-dimensional methods: Newton's method (cont'd)

L09: convergence of Newton and its enhancements

Newton equation at $\mathbf{x}^{(k)}$:

$$\mathbf{H}_f^{(k)}\mathbf{d}^{(k)} = -\mathbf{g}^{(k)} \quad \Longrightarrow \quad \text{Newton iteration : } \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{d}^{(k)}.$$

- **Local Q-quadratic rate**: suppose $\nabla f(\mathbf{x}^*) = \mathbf{0}$, $\mathbf{H}_f(\mathbf{x}^*)$ is nonsingular, and the Hessian is locally Lipschitz. For a sufficiently close start,
$$\|\mathbf{x}^{(k+1)} - \mathbf{x}^*\| \leq C\|\mathbf{x}^{(k)} - \mathbf{x}^*\|^2.$$
- **One-step termination**: if f is quadratic and $\mathbf{H}_f$ is nonsingular, Newton converges in at most one step.
- Enhancements:
  - Well-posedness: choose a **shift** so that $\mathbf{H}_f^{(k)} + \tau^{(k)}\mathbf{I} \succ 0$.
  - Robustness: enforce descent direction and **Armijo backtracking**

Focus. Local Q-quadratic rate under assumptions, Armijo backtracking.

Multi-dimensional methods: Newton's method (cont'd)

L09: convergence of Newton and its enhancements

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Focus. Local Q-quadratic rate under assumptions, Armijo backtracking.

Mistake. (1) Claiming quadratic/one-step convergence in any case.
(2) Equating Armijo backtracking with exact line search or a full step.

Multi-dimensional methods: steepest descent

L04, L10: steepest descent from a more naive local quadratic model

At $\mathbf{x}^{(k)}$, let $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$. With a fixed $\eta > 0$, consider

$$m_{\mathrm{SD}}^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2\eta}\|\mathbf{d}\|^2 \quad \Longrightarrow \quad \mathbf{d}^{(k)} = -\eta \mathbf{g}^{(k)}.$$

The algorithm uses the same ray:

$$\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}, \qquad \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha^{(k)} \mathbf{g}^{(k)}.$$

- Step size $\alpha^{(k)}$: exact or inexact (e.g., Armijo) line search.
- For a PD quadratic, exact line search gives

$$\alpha^{(k)} = \frac{\|\mathbf{g}^{(k)}\|^2}{(\mathbf{g}^{(k)})^T \mathbf{Q} \mathbf{g}^{(k)}} \qquad (\mathbf{g}^{(k)} \neq \mathbf{0}).$$

Focus. Steepest descent direction.

Multi-dimensional methods: steepest descent

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Focus. Steepest descent direction.

Mistake. (1) Stepping along $+\mathbf{g}^{(k)}$. (2) Using an exact-search formula when the specified search is inexact.

Multi-dimensional methods: steepest descent (cont'd)

$$\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}, \qquad \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha^{(k)}\mathbf{g}^{(k)}.$$

- **Global error estimate**: assume f is μ-strongly convex and L-smooth ($0 < \mu \leq L$), with $\alpha^{(k)} \equiv 1/L$ or by exact line search,

$$f(\mathbf{x}^{(k)}) - f^* \leq \left(1 - \frac{\mu}{L}\right)^k \left(f(\mathbf{x}^{(0)}) - f^*\right).$$

- **Condition number**: $\kappa = L/\mu$. For a PD quadratic, $L = \lambda_{\max}(\mathbf{Q})$, $\mu = \lambda_{\min}(\mathbf{Q})$.
- **Zig-zagging** by exact line search:

$$(\mathbf{g}^{(k+1)})^T \mathbf{g}^{(k)} = 0.$$

- Enhancement: **preconditioned** steepest descent.

Focus. Global Q-linear rate under assumptions, condition number.

Multi-dimensional methods: steepest descent (cont'd)

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- Enhancement: **preconditioned** steepest descent.

Focus. Global Q-linear rate under assumptions, condition number.

Mistake. (1) Claiming the linear rate without strong convexity and smoothness. (2) Reversing the eigenvalue ratio in κ.

Multi-dimensional methods: conjugate direction method

L11: conjugate directions from a PD quadratic and exact line searches

Let $q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}$, $\mathbf{Q} \succ 0$. Nonzero directions are **Q-conjugate** when

$$(\mathbf{d}^{(i)})^T\mathbf{Q}\mathbf{d}^{(j)} = 0 \quad (i \neq j).$$

Along the full line ($\alpha \in \mathbf{R}$), minimize $q(\mathbf{x}^{(k)} + \alpha\mathbf{d}^{(k)})$,

$$\alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^T\mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T\mathbf{Q}\mathbf{d}^{(k)}}, \qquad \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)}\mathbf{d}^{(k)}.$$

Memory property: the iterate after k steps minimizes q over

$$\mathbf{x}^{(0)} + \text{span}\{\mathbf{d}^{(0)}, \ldots, \mathbf{d}^{(k-1)}\}.$$

Focus. **Q**-conjugacy, finite termination.

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Focus. **Q**-conjugacy, finite termination.

Mistake. (1) Equating **Q**-conjugacy with ordinary orthogonality. (2) Applying finite termination without $\mathbf{Q} \succ 0$ and exact arithmetic.

Multi-dimensional methods: linear conjugate-gradient

L11: conjugate gradient as a special instance of conjugate direction

Initialize $\mathbf{g}^{(0)} = \mathbf{Q}\mathbf{x}^{(0)} - \mathbf{c}$, $\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}$. For $\mathbf{g}^{(k)} \neq \mathbf{0}$, compute

$$
\begin{aligned}
\alpha^{(k)} &= \frac{\|\mathbf{g}^{(k)}\|^2}{(\mathbf{d}^{(k)})^T \mathbf{Q}\mathbf{d}^{(k)}}, \\
\mathbf{x}^{(k+1)} &= \mathbf{x}^{(k)} + \alpha^{(k)}\mathbf{d}^{(k)}, \\
\mathbf{g}^{(k+1)} &= \mathbf{g}^{(k)} + \alpha^{(k)}\mathbf{Q}\mathbf{d}^{(k)}, \\
\beta^{(k)} &= \frac{\|\mathbf{g}^{(k+1)}\|^2}{\|\mathbf{g}^{(k)}\|^2}, \\
\mathbf{d}^{(k+1)} &= -\mathbf{g}^{(k+1)} + \beta^{(k)}\mathbf{d}^{(k)}.
\end{aligned}
$$

Stop when $\mathbf{g}^{(k+1)} = \mathbf{0}$, before forming the next direction.

Focus. Direction update, conjugate parameter $\beta^{(k)}$, step size $\alpha^{(k)}$.

Multi-dimensional methods: linear conjugate-gradient

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\end{aligned}
$$

Stop when $\mathbf{g}^{(k+1)} = \mathbf{0}$, before forming the next direction.

Focus. Direction update, conjugate parameter $\beta^{(k)}$, step size $\alpha^{(k)}$.

Mistake. Mixing gradient/residual signs, or substituting $\mathbf{g}^{(k)}$ for $\mathbf{g}^{(k+1)}$ in the direction update.

Multi-dimensional methods: linear CG (cont'd)

L11: convergence of linear conjugate gradient

For exact linear CG on the same PD quadratic, define

$$\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}^*, \qquad \|\mathbf{e}\|_{\mathbf{Q}} = \sqrt{\mathbf{e}^T \mathbf{Q} \mathbf{e}}, \qquad \kappa = \frac{\lambda_{\max}(\mathbf{Q})}{\lambda_{\min}(\mathbf{Q})}.$$

In exact arithmetic:

- **Finite termination:** at most n updates in $\mathbf{R}^n$; at most one if $\kappa = 1$.
- **Global error estimate:** for $\kappa > 1$,

$$\|\mathbf{e}^{(k)}\|_{\mathbf{Q}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \|\mathbf{e}^{(0)}\|_{\mathbf{Q}}.$$

- Variants: preconditioned CG, nonlinear CG.

Focus. Finite termination, convergence rate in $\sqrt{\kappa}$.

Multi-dimensional methods: linear CG (cont'd)

L11: convergence of linear conjugate gradient

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- Variants: preconditioned CG, nonlinear CG.

Focus. Finite termination, convergence rate in $\sqrt{\kappa}$.

Mistake. Extending exact quadratic-CG guarantees to indefinite problems, nonlinear CG, or floating-point termination.

Compare methods under the same assumptions

L09–L11: one PD quadratic in exact arithmetic

Method	Main iteration work	Main guarantee
Newton	Solve $\mathbf{Qd} = -\mathbf{g}$	One full step
Steepest descent	Mat–vec product and inner products	κ-dependent bound
Linear CG	Mat–vec product, inner products, extra vector updates	$\sqrt{\kappa}$-dependent bound; at most n steps

For general nonlinear objectives and practical computing environments,

- additional enhancements (adaptive direction, step size, ...);
- round-off errors due to the finite precision;
- additional regularity assumptions.

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For general nonlinear objectives and practical computing environments,

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- round-off errors due to the finite precision;
- additional regularity assumptions.

Mistake. Comparing performance using iteration counts alone.

Choose the right reasoning for the question

Question	What a complete answer checks
Does a minimizer exist?	Nonempty domain, attainment theorem, or an explicit optimizer.
Is this point optimal?	Feasibility; the applicable necessary or sufficient conditions; local versus global conclusion.
Perform an iteration	Current/previous derivative/residual; well-posedness; direction; step rule.
Does the method converge?	Problem class, initialization, step assumptions, error quantity, and meaning of the rate.

Self-check: true or false?

1. A finite optimal value implies that an optimizer exists.
2. A stationary point with PSD Hessian must be a local minimum.
3. Nonnegative leading principal minors guarantee that a symmetric matrix is positive semidefinite.
4. If $\mathbf{Q} \succeq 0$ and $\mathbf{c} \in \text{Range}(\mathbf{Q})$, then $\frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}$ attains its minimum.
5. A PD Hessian guarantees that the full Newton step decreases the objective.
6. Exact linear CG on an n-dimensional PD quadratic always needs exactly n iterations.

Self-check answers

1. **False.** e^x on $\mathbf{R}$ has finite infimum 0, never attained.
2. **False.** $-x^4$ at 0 has zero first and second derivatives, but is a maximum.
3. **False.** $\mathrm{diag}(0, -1)$ has nonnegative leading principal minors but a negative eigenvalue.
4. **True.** Choose $\mathbf{Q}\mathbf{x}_0 = \mathbf{c}$. Then $q(\mathbf{x}) - q(\mathbf{x}_0) = \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T \mathbf{Q}(\mathbf{x} - \mathbf{x}_0) \geq 0$.
5. **False.** For $f(x) = e^x - x$, the full Newton step from $x = -2$ gives $x^+ = e^2 - 3$ and increases f, despite $f'' > 0$.
6. **False.** The bound is *at most* n. Earlier termination is possible.